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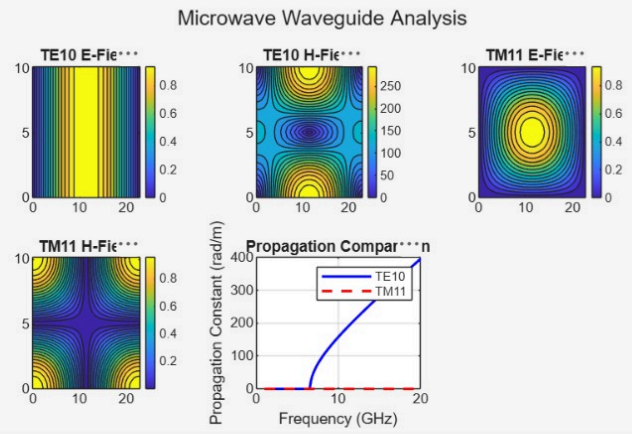
m1.m

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```
clc; clear; close all;
c=3e8; f=10e9; a=22.86e-3; b=10.16e-3;
fc1=c/(2*a); fc2=c/2*sqrt((1/a)^2+(1/b)^2);
[x,y]=meshgrid(linspace(0,a,40),linspace(0,b,25));
E1=sin(pi*x/a); H1=abs(cos(pi*x/a).*cos(pi*y/b));
E2=sin(pi*x/a).*sin(pi*y/b);
H2=sqrt(((pi/a)*cos(pi*x/a).*sin(pi*y/b)).^2+((pi/b)*sin(pi*x/a).*cos
figure;
subplot(2,3,1); contourf(x*1e3,y*1e3,E1,15); title('TE10 E-Field');
subplot(2,3,2); contourf(x*1e3,y*1e3,H2,20); title('TE10 H-Field');
subplot(2,3,3); contourf(x*1e3,y*1e3,E2,15); title('TM11 E-Field');
subplot(2,3,4); contourf(x*1e3,y*1e3,H1,20); title('TM11 H-Field');
q=1:0.1:20; beta1=zeros(size(q)); i=q>=fc1/1e9;
beta1(i)=sqrt((2*pi*q(i)*1e9/c).^2-(pi/a)^2); beta2=zeros(size(q));
subplot(2,3,5); plot(q,beta1,'b','LineWidth',1.5); hold on; plot(q,be
title('Propagation Comparison'); xlabel('Frequency (GHz)'); ylabel('P
sgtitle('Microwave Waveguide Analysis');
```

Apps

Figure 1



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